

Chapter 2

How Time and Interest affect money

In Chapter 1, we learned that money has a time value and that interest can be used to move money back and forth across a timeline. However, solving these fundamental equations repeatedly for complex series of payments is tedious and error-prone.

To simplify these calculations, engineers use mathematical multipliers known as **engineering economy factors**. A factor is simply a pre-calculated formula that directly converts one type of cash flow (like a present amount) into another (like a future amount or an annual series) at a given interest rate.

This chapter develops the derivations for all the commonly used factors that take the time value of money into account, providing a standardized and efficient toolkit for evaluating any cash flow scenario.

2.1 Notation

Before delving into this Chapter, let us define the notation we will use:

- P : Present Worth
- F : Future Worth
- A : Annual Amount
- n : number of interest periods
- i : interest rate (or rate of return) per period

- G : Arithmetic Gradient
- g : Geometric Gradient

2.2 Single-Amount Factors

2.2.1 Single-Payment Compound Amount

Recall that P dollars now are equivalent to F dollar after n time periods at an interest rate of i per time period¹, where

$$F = P(1 + i)^n$$

Definition – Single-Payment Compound Amount Factor (F/P)

The Single-Payment Compound Amount factor $(1 + i)^n$ is the F/P factor. The standard notation for the F/P factor is

$$(F/P, i, n) = (1 + i)^n$$

Example 2.1. If you deposit \$1,000 today in a savings account that pays 5% interest per year, how much will you have in 4 years?

$$F = P(F/P, 5\%, 4) = 1000(1 + 0.05)^4 = \$1,215.51$$

2.2.2 Single-Payment Present Worth

Alternatively, if we have the Future Worth (F) and we want to find the Present Worth (P):

$$P = \frac{F}{(1 + i)^n} = F(1 + i)^{-n}$$

Definition – Single-Payment Present Worth Factor (P/F)

The Single-Payment Present Worth factor $(1 + i)^{-n}$ is the P/F factor. The standard notation for the P/F factor is

$$(P/F, i, n) = \frac{1}{(1 + i)^n} = (1 + i)^{-n}$$

¹When the type of interest is not specified, assume compound interest

Example 2.2. If you want to have \$5,000 in 3 years to buy a car, and the interest rate is 4% per year, how much should you deposit today?

$$P = F(P/F, 4\%, 3) = 5000(1 + 0.04)^{-3} = \$4,444.98$$

Important – Standard Factor Notation

A standard notation has been adopted for all factors. It is in the form $(X/Y, i, n)$, which reads: find X given Y where i is the interest rate per period and n is the number of periods.

These two factors, as well as other factors, are tabulated at the end of this chapter. Spreadsheets (i.e., Excel) has built-in function for factors. Excel is very practical for such calculations. Table 2.1 summarizes these two factors, and the corresponding formulas and functions.

Note – Excel FV Function Sign Convention

When working in Excel, the computation depends on the definition of the factor. For instance, the FV function “calculates the future value of an investment based on a constant interest rate”. As a result, the investment is a payment and should be a negative amount.

Notation	Name	Equation	Excel
$(F/P, i, n)$	Compound Amount Factor	$F = P(1 + i)^n$	$FV(i\%, n, ^2, P)$
$(P/F, i, n)$	Present Worth Factor	$P = F(1 + i)^{-n}$	$PV(i\%, n, , F)$

Table 2.1: P/F and F/P factors: notations, equation, and excel

2.3 Uniform Series Factors

2.3.1 Uniform-Series Present Worth Factor (P/A) and Uniform-Series Compound Amount Factor (F/A)

Example 2.3. Dieter wants to save money for his newborn child. He decides to invest \$1200 every year until his daughter, Bertha, is 18 years old. The interest rate on the investments is 3% yearly.

What is the Future Worth of this investment when Bertha is 18 years old?

We begin first by finding the Present Worth of the investment. To this end, we need to find the equivalent present worth of each Annual amount A . To understand the logic,

²Note that this payment term (pmt) is not used

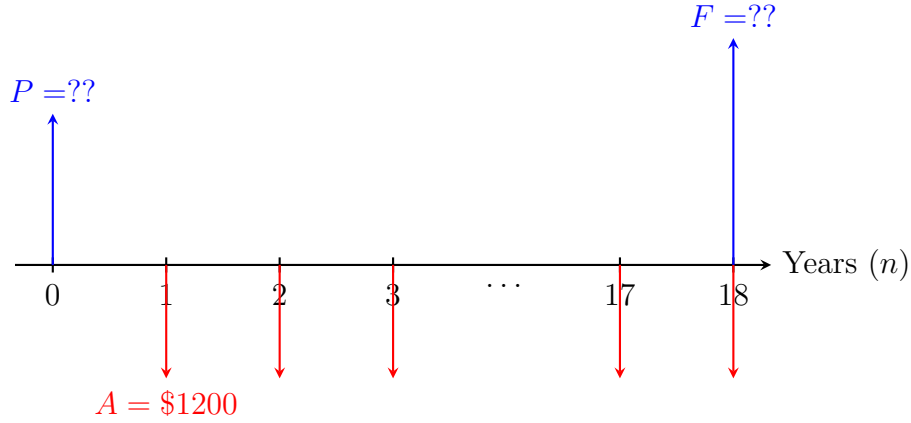


Figure 2.1: Cash flow diagram for Bertha's savings plan. Note that the present worth is allocated before the first investment and the future worth is allocated at the same time as the last investment.

consider a simple case with $n = 3$ periods:

$$P = \frac{A}{1+i} + \frac{A}{(1+i)^2} + \frac{A}{(1+i)^3}$$

We can generalize this derivation for n periods:

$$\begin{aligned}
 P &= \underbrace{\frac{A}{1+i}}_{1^{st} \text{ period equivalence}} + \underbrace{\frac{A}{(1+i)^2}}_{2^{nd} \text{ period equivalence}} + \cdots + \frac{A}{(1+i)^n} \\
 &= \frac{A}{1+i} \left(1 + \frac{1}{1+i} + \cdots + \left(\frac{1}{1+i} \right)^{n-1} \right) \\
 &= \frac{A}{1+i} \sum_{j=0}^{n-1} \left(\frac{1}{1+i} \right)^j \quad \text{Given that } \sum_{j=0}^{n-1} x^j = \frac{x^n - 1}{x - 1} \\
 &= \frac{A}{1+i} \left(\frac{\left(\frac{1}{1+i} \right)^n - 1}{\left(\frac{1}{1+i} \right) - 1} \right) \\
 &= A \left(\frac{\left(\frac{1}{1+i} \right)^n - 1}{1 - (1+i)} \right) \quad \text{multiply both numerator and denominator by } (1+i)^n \\
 &= A \left(\frac{1 - (1+i)^n}{-i(1+i)^n} \right) \\
 &= A \left(\frac{(1+i)^n - 1}{i(1+i)^n} \right) \\
 &= A \times (P/A, i, n)
 \end{aligned}$$

Replacing n by 18, we will get that the Present Worth of the investment is:

$$P = 1200 \left(\frac{(1 + 0.03)^{18} - 1}{0.03(1 + 0.03)^{18}} \right) = 16,504.22$$

To find the Future Worth, we use the F/P factor:

$$F = P(F/P, i, n) = 16,504.22(1 + 0.03)^{18} = 28,097.32$$

To find the Future Worth, given the annuities, we compute the F/A factor as follows:

$$\begin{aligned} F/A &= F/P \times P/A \\ &= (1 + i)^n \times \left(\frac{(1 + i)^n - 1}{i(1 + i)^n} \right) \\ &= \frac{(1 + i)^n - 1}{i} \end{aligned}$$

Applying this formula to Bertha's savings:

$$F = A \frac{(1 + i)^n - 1}{i} = 1200 \frac{(1.03)^{18} - 1}{0.03} = 28,097.32$$

Definition – Uniform Series Factors (P/A and F/A)

The P/A factor is defined as:

$$\left(\frac{(1 + i)^n - 1}{i(1 + i)^n} \right)$$

The F/A factor is defined as:

$$\frac{(1 + i)^n - 1}{i}$$

Table 2.2 summarizes the notation and the Excel formulas.

Notation	Name	Equation	Excel
$(P/A, i, n)$	Uniform series present worth	$\left(\frac{(1+i)^n - 1}{i(1+i)^n} \right)$	$FV(i\%, n, A)$
$(F/A, i, n)$	Uniform series compound amount	$\left(\frac{(1+i)^n - 1}{i} \right)$	$FV(i, n, A)$

Table 2.2: P/A and F/A factors: notations, equation, and excel

Important – Timing of Present Worth

As shown in Figure 2.1, the Present Worth should always be allocated before the first payment.

Important – Timing of Future Worth

As shown in Figure 2.1, the Future Worth and the last payment occur at the same time.

2.3.2 Capital Recovery Factor (A/P) and Sinking Fund Factor (A/F)

In some situation (i.e., car loan), we need to find the payments over a number of periods. Given that we have the P/A and F/A factors, finding the A/P and A/F is straightforward.

Definition – Capital Recovery Factor (A/P)

The A/P factor is defined as:

$$\frac{i(1+i)^n}{(1+i)^n - 1}$$

Definition – Sinking Fund Factor (A/F)

The A/F factor is defined as:

$$\frac{i}{(1+i)^n - 1}$$

Notation	Name	Equation	Excel
$(A/P, i, n)$	Capital recovery	$\left(\frac{i(1+i)^n}{(1+i)^n - 1} \right)$	$PMT(i\%, n, P)$
$(A/F, i, n)$	Sinking fund	$\left(\frac{i}{(1+i)^n - 1} \right)$	$PMT(i, n, F)$

Table 2.3: A/P and A/F factors: notations, equation, and excel

Example 2.4. Bart wants to buy a Nissan GT-R R34 for 200,000. He has saved a down payment of 80,000 and plans to finance the rest using a 5-year auto loan. The bank offers him an interest rate of 7% **per year**, compounded **monthly**.

1. What is Bart's required monthly payment?
2. What is the total out-of-pocket cash Bart will pay to the bank over the 5 years?
3. What is the true Future Worth (F) of this loan at the end of the 5 years?

First, we must determine the principal amount borrowed (P) and ensure our interest rate (i) and number of periods (n) match the monthly payment schedule.

- Principal (P) = $200000 - 80000 = 120000$
- Monthly Interest Rate (i) = $7/12 = 0.5833\%$ per month³
- Number of Periods (n) = $5 \times 12 = 60$

To find the uniform monthly payment, we use the Capital Recovery Factor (A/P):

$$\begin{aligned}
 A &= P(A/P, i, n) \\
 &= 120000 \left(\frac{0.005833(1 + 0.005833)^{60}}{(1 + 0.005833)^{60} - 1} \right) \\
 &= 120000(0.01980) \\
 &= 2,376.14 \text{ per month}
 \end{aligned}$$

Note – Excel PMT Function

Note: In Excel, this can be computed directly as ‘=PMT(0.07/12, 60, -120000)’. Notice that the principal is entered as a negative value so the function returns a positive payment amount.

The total out-of-pocket cash paid is:

$$2376.14 \times 60 = 142,568.40$$

The Future Worth of the payments is:

$$F = A(F/A, i, n) = 170,115.03$$

Cashflow Diagram

In this Section, we show the cashflow diagram of the car loan example. We use this cashflow to re-highlight key features of a cashflow diagram.

Key Features

³Here we simply divide the annual rate by 12 because 7% is a nominal rate. We will formally discuss the difference between nominal and effective interest rates in the next chapter.

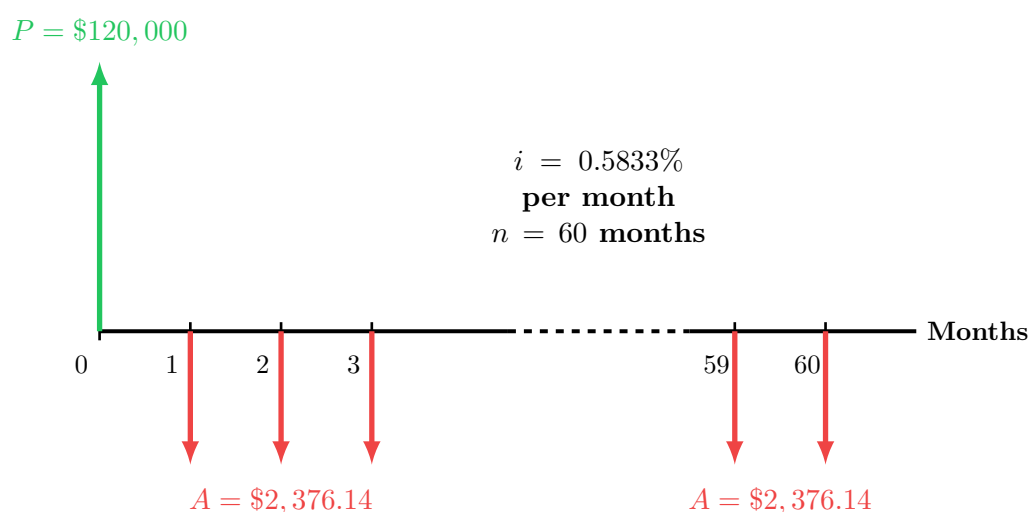


Figure 2.2: Cash flow diagram for a 60-month GT-R R34 car loan from the borrower's perspective. The upward arrow at time zero represents the principal amount P received, while the uniform downward arrows represent the equal monthly repayments A over the n -month loan term.

- The horizontal axis represents the time, where each period is a month.
- The upward arrow (Green) represents the cash inflow, and is a positive value as viewed by the borrower.
- The downward arrows (in red) represent the monthly payments that the borrower should pay.
- The payments are paid at the end of each period.
- The present value that the borrower receives occurs at time 0, before paying the first payment.
- If we are to illustrate the Future Worth of the loan, it would coincide at period 60 together with the last payment.

2.4 Running Amortization

When Bart took a \$120,000 loan and starts paying back the bank \$2,376.14 per month, the bank view this **amortization payment** A as composed of two parts: (1) interest on running (outstanding) balance and (2) partial repayment of the principal.

Definition – Amortizing Loan

An amortizing loan is a loan where the principal of the loan is paid down over the life of the loan (that is, amortized) according to an amortization schedule, typically through equal payments.

source: Wikipedia

To make this concept more concrete, let us revisit the car loan in details.

At the end of the first period, the interest on the first period is $120,000 \times 0.005833 = 699.96$. Then, 699.96 is used to pay for the interest of the first period, while the remaining 1676.18 is used to pay the principal. At the end of this period, and after paying the payment, the new balance is the (original principal + interest - payment). Figure 2.3 illustrates how the payment is used to partially cover the principal and interest over time.

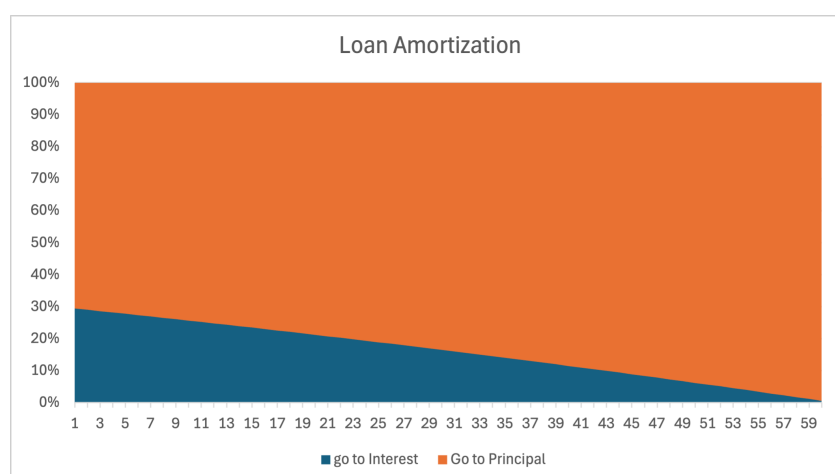


Figure 2.3: The proportion of payment to repay the principal vs repaying the interest over the loan period

End of Period	Previous Balance	Payment (A)	Interest	Principal Paid	New Balance
0	—	—	—	—	\$120,000.00
1	\$120,000.00	\$2,376.14	\$700	\$1,676.14	\$118,323.85
2	\$118,323.82	\$2,376.14	\$690.18	\$1,685.96	\$116,637.86
3	\$116,637.86	\$2,376.14	\$680.35	\$1,695.79	\$114,942.07
⋮	⋮	⋮	⋮	⋮	⋮
59	\$4,709.69	\$2,376.14	\$27.48	\$2,348.66	\$2,362.36
60	\$2,362.36	\$2,376.14	\$13.78	\$2,362.36	≈ 0

Table 2.4: Running amortization schedule for the 60-month GT-R R34 car loan. While the last New Balance should be exactly equal to 0, this small error comes from the accumulation of rounding errors.

2.5 Arithmetic Gradient Factors (P/G) and (A/G)

In many scenarios, the periodic payment may not be constant, and could change over the course of the time, as illustrate in Figure 2.4. In this particular illustration, the changes are constant. We call this kind of series **arithmetic gradient**.

Definition – Arithmetic Gradient Series

An **arithmetic gradient** series is a cash flow series that either increases or decreases by a **constant amount** each period.

The amount of change is called the **gradient**.

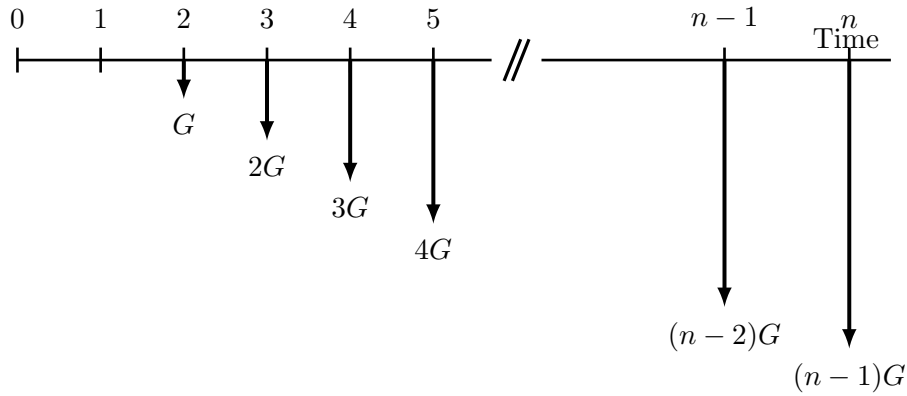


Figure 2.4: Cash flow diagram for an arithmetic gradient series

In the general case, the cash flow CF_n at point n consists of a base amount and a gradient as shown in Figure 2.5:

$$CF_n = \text{base amount} + (n - 1)G$$

2.5.1 Present Worth

At time 0, the present worth is the sum of the two series:

$$P = P_A + P_G$$

We know $P_A = A(P/A, i, n)$. To find P_G , we sum the present worth of each gradient component:

$$P_G = \frac{G}{(1+i)^2} + \frac{2G}{(1+i)^3} + \cdots + \frac{(n-1)G}{(1+i)^n}$$

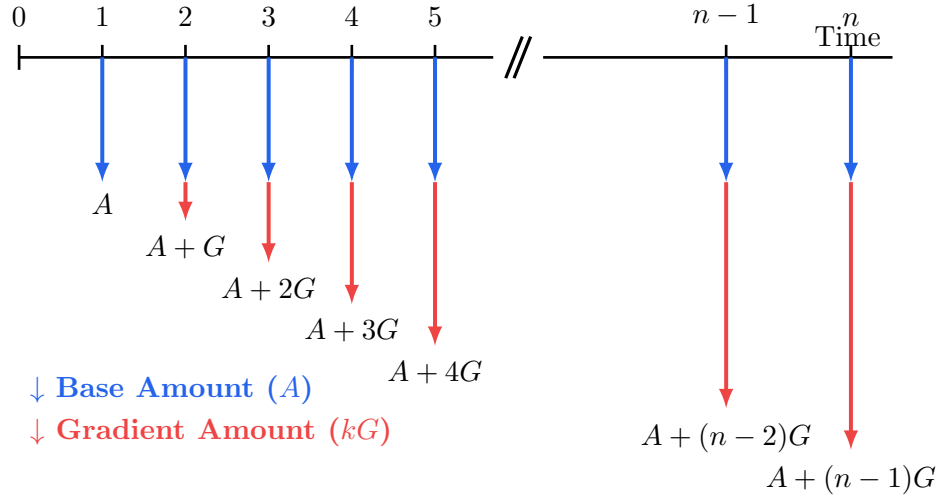


Figure 2.5: Cash flow diagram illustrating a composite arithmetic gradient series, showing the uniform base amount (A) and the increasing gradient component (G).

$$= \frac{G}{1+i} \sum_{j=1}^{n-1} \left(\frac{j}{(1+i)^j} \right)$$

This summation forms an arithmetico-geometric sequence. Resolving this standard summation and simplifying yields the final formula:

$$\begin{aligned} P_G &= \frac{G}{1+i} \left(\frac{(1+i)^{1-n} ((1+i)^n - ni - 1)}{i^2} \right) \\ &= G \left(\frac{(1+i)^n - ni - 1}{i^2(1+i)^n} \right) \end{aligned}$$

Definition – Arithmetic Gradient Present Worth Factor (P/G)

The P/G factor is:

$$(P/G, i, n) = \left(\frac{(1+i)^n - ni - 1}{i^2(1+i)^n} \right)$$

2.5.2 Equivalent uniform series

Finding the equivalent uniform series A_G for an arithmetic gradient series requires finding the arithmetic gradient uniform series factor A/G :

$$A/G = A/P \times P/G$$

As a result,

$$\begin{aligned} A_G &= G(A/P, i, n)(P/G, i, n) \\ &= G\left(\frac{1}{i} - \frac{n}{(1+i)^n - 1}\right) \end{aligned}$$

Definition – Arithmetic Gradient Uniform Series Factor (A/G)

The arithmetic gradient uniform series factor A/G factor is:

$$\left(\frac{1}{i} - \frac{n}{(1+i)^n - 1}\right)$$

2.5.3 Future Worth

Finding F_G is done similar to the previous factor:

$$\begin{aligned} F_G &= G(F/P, i, n)(P/G, i, n) \\ &= G\left(\frac{(1+i)^n - in - 1}{i^2}\right) \end{aligned}$$

Example 2.5. A manufacturing plant installs a new cobot. The maintenance costs are expected to be 1,000 at the end of the first year and will increase by 200 each subsequent year for a total of 5 years. If the company's Minimum Attractive Rate of Return (MARR) is 8%, what is (1) the equivalent present worth (P) and (2) the equivalent uniform annual cost (A) of this maintenance?

We begin by showing the cashflow diagram in Figure 2.6. The present worth of the

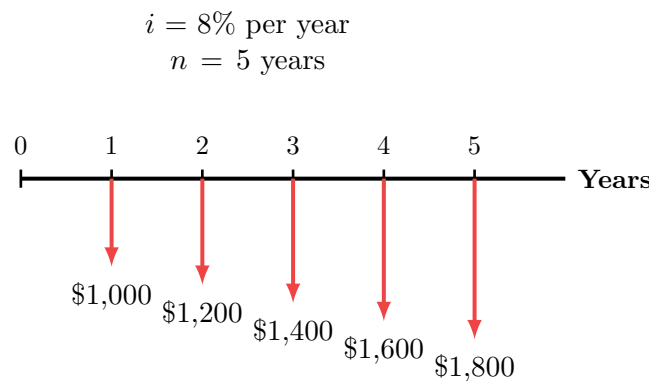


Figure 2.6: Cash flow diagram for the robotic arm maintenance costs, illustrating a composite series with a base amount of \$1,000 and an arithmetic gradient of \$200.

two series is:

$$\begin{aligned}
 P &= P_A + P_G \\
 &= A(P/A, 8\%, 5) + G(P/G, 8\%, 5) \\
 &= A \left(\frac{(1+i)^n - 1}{i(1+i)^n} \right) + G \left(\frac{(1+i)^n - ni - 1}{i^2(1+i)^n} \right) \\
 &= 3992.71 + 1474.48 \\
 &= 5467.19
 \end{aligned}$$

The equivalent uniform annual cost for this maintenance schema is:

$$\begin{aligned}
 A_{\text{equiv}} &= P(A/P, 8\%, 5) \\
 &= 5467.19 \left(\frac{0.08(1+0.08)^5}{(1+0.08)^5 - 1} \right) \\
 &= 1369.29
 \end{aligned}$$

While the **arithmetic** gradient series increases (or decrease) in a fixed amount every period, some series exhibit a different more exponential behaviour.

2.6 Geometric Gradient Factors

In many situations, the changes in the periodic revenues or costs change by a **constant** percentage.

Example 2.6. Suppose a machine requires a specialized lubricant. At the end of Year 1, the cost of the lubricant is \$100. Due to inflation, the supplier announces that the price will increase by exactly 10% every year.

What are the actual cash flows for the first four years?

Solution:

We start with a base amount of $A_1 = \$100$ and a growth rate of $g = 0.10$. Every subsequent year is calculated by multiplying the previous year by $(1 + g)$.

- **Year 1 (A_1): \$100**
- **Year 2:** The previous year plus 10%.

$$\$100 \times 1.10 = \$110$$

(An increase of \$10)

- **Year 3:** The previous year plus 10%.

$$\$110 \times 1.10 = \$121$$

(An increase of \$11)

- **Year 4:** The previous year plus 10%.

$$\$121 \times 1.10 = \$133.1$$

(An increase of \$12.1)

Notice that the amount added each year (\$10, then \$11, then \$12.10) is not constant. Because the 10% growth rate is applied to a slightly larger base every year, the cash flow curve bends upwards exponentially.

Definition – Geometric Gradient Series

A **geometric gradient** series is a cash flow series that either increases or decreases by a **constant percentage** each period.

The uniform change is called the **rate of change**.

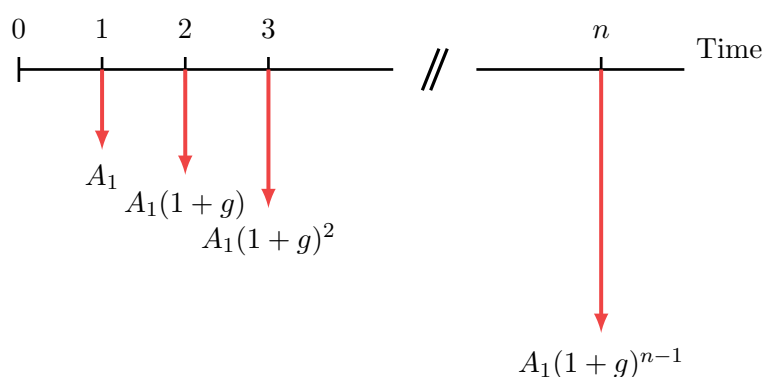


Figure 2.7: General cash flow diagram for a geometric gradient series extending to n periods.

As shown in Figure 2.7, the series is defined by an initial cash flow A_1 in period 1, followed by a constant increase of rate g . Let

- g = **constant rate of change**, in decimal form, by which cash flow values increase or decrease from one period to the next. The gradient g can be $+$ or $-$
- A_1 = initial cash flow in period 1 of the geometric series

- P_g = **present worth** of the entire geometric gradient series, including the initial amount A_1

Proof – Geometric Gradient Present Worth

The Present worth of the series is:

$$\begin{aligned} P_g &= \frac{A_1}{1+i} + \frac{A_1(1+g)}{(1+i)^2} + \cdots + \frac{A_1(1+g)^{n-1}}{(1+i)^{n-1}} + \frac{A_1(1+g)^n}{(1+i)^n} \\ &= \frac{A_1}{1+i} \sum_{j=0}^{n-1} \left(\frac{1+g}{1+i} \right)^j \end{aligned} \quad (2.1)$$

The geometric series $\sum_{j=0}^{n-1} \left(\frac{1+g}{1+i} \right)^j$ is equivalent to:

$$\sum_{j=0}^{n-1} \left(\frac{1+g}{1+i} \right)^j = \frac{\left(\frac{1+g}{1+i} \right)^n - 1}{\left(\frac{1+g}{1+i} \right) - 1}$$

Substituting back in Equation 2.1:

$$\begin{aligned} P_g &= \frac{A_1}{1+i} \sum_{j=0}^{n-1} \left(\frac{1+g}{1+i} \right)^j \\ &= \frac{A_1}{1+i} \frac{\left(\frac{1+g}{1+i} \right)^n - 1}{\left(\frac{1+g}{1+i} \right) - 1} \\ &= A_1 \frac{\left(\frac{1+g}{1+i} \right)^n - 1}{g - i} \end{aligned}$$

The Present worth of the series is:

$$P_g = A_1 \frac{\left(\frac{1+g}{1+i} \right)^n - 1}{g - i}$$

When the interest is equivalent to the rate of change, that is $i = g$, the standard formula would result in a division by zero. Instead, looking back at the original summation (Equation 2.1), if $g = i$, the fraction $\frac{1+g}{1+i}$ simply becomes 1. Summing 1 for n periods simply yields n . Thus, the Present Worth formula simplifies beautifully to:

$$P_g = \frac{nA_1}{1+i}$$

Definition – Geometric Gradient Factor ($P/A, g, i, n$)

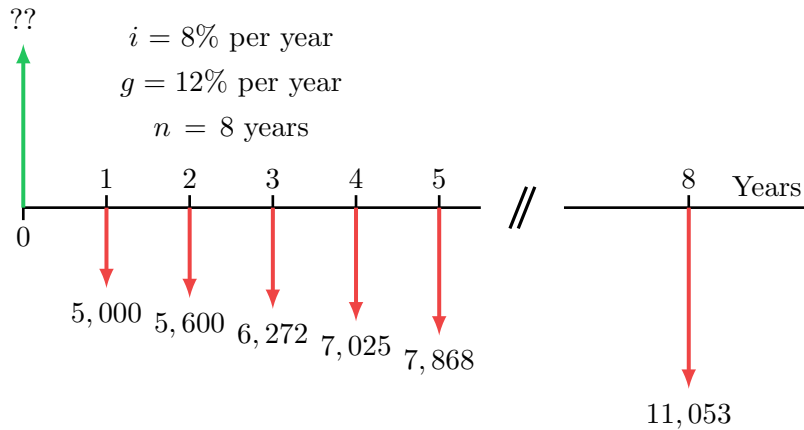
The $(P/A, g, i, n)$ factor calculates P_g in period $t = 0$ for a geometric gradient series starting in period 1 in the amount A_1 and increasing by a constant rate of g each period.

$$(P/A, g, i, n) = \begin{cases} \frac{\left(\frac{1+g}{1+i}\right)^n - 1}{g-i} & g \neq i \\ \frac{n}{1+i} & g = i \end{cases}$$

Example 2.7. The Lord of the Rings: Mining in Moria

The Dwarves of Khazad-dûm (Moria) are expanding their Mithril mining operations. Gimli is reviewing the cost of reinforcing the deep mining shafts. The deeper they dig, the more complex and dangerous the operation becomes, leading to costs that increase geometrically. The cost to reinforce the new shafts at the end of the first year (A_1) is estimated at 5,000 gold pieces. Because the rock becomes more unstable and the threat of awakening ancient evils increases, this cost is expected to grow by 12% each year ($g = 12\%$). The Iron Hills Bank offers a lending rate of 8% per year ($i = 8\%$).

What is the equivalent present worth (P_g) of the shaft reinforcement costs over an 8-year expansion plan?



Solution: We are given:

- $A_1 = 5,000$
- $g = 0.12$
- $i = 0.08$
- $n = 8$

Since $i \neq g$, we use the general formula for the present worth of a geometric gradient series:

$$P_g = A_1 \left(\frac{\left(\frac{1+g}{1+i} \right)^n - 1}{g - i} \right)$$

Substituting the given values:

$$\begin{aligned} P_g &= 5,000 \left(\frac{\left(\frac{1+0.12}{1+0.08} \right)^8 - 1}{0.12 - 0.08} \right) \\ &= 5,000 \left(\frac{\left(\frac{1.12}{1.08} \right)^8 - 1}{0.04} \right) \\ &= 5,000 \left(\frac{(1.037037)^8 - 1}{0.04} \right) \\ &= 5,000 \left(\frac{1.3377 - 1}{0.04} \right) \\ &= 5,000 (8.442) \\ &\approx 42,210 \text{ gold pieces} \end{aligned}$$

Therefore, the present worth of the reinforcement costs over the next 8 years is approximately 42,275 gold pieces.

Important – Cash Flow Timing and Present Worth Location

Note that the Present Worth is in period 0 before the initial amount.

The time stamp of the cashflow affect the derivation of the factors. Hence, it is **important** to know the cash flow diagram to properly understand the time stamp of the various cash flows.

2.7 Summary

This chapter introduced the **foundational** engineering economy factors used to calculate the time value of money. We began with single-amount factors (F/P and P/F) before extending the logic to uniform series (P/A , F/A , A/P , and A/F) and gradients.

A key element in applying these factors correctly is to properly illustrate the cash flows using a cash flow diagram. This visualization is especially crucial when dealing with complex, composite cash flows. By drawing a clear diagram, we can conceptually decompose any complicated problem into simpler parts that can be solved using the standard factors derived in this chapter (See Example 2.8).

It is worth noting that the same time-value-of-money relationships can be rearranged to solve for **unknown variables** other than P , F , or A . For example, given sufficient information about the cash flows, one can determine:

- **The interest rate i :** For single amounts, the $F = P(1 + i)^n$ relation can be solved directly as $i = (F/P)^{1/n} - 1$. For uniform series, however, the factor equations are nonlinear in i and generally require iterative methods (e.g., trial-and-error with linear interpolation between tabulated factor values). In practice, spreadsheet functions such as `=RATE(n, A, P, F)` handle this computation directly.
- **The number of periods n :** For single amounts, taking logarithms yields $n = \ln(F/P) / \ln(1 + i)$. For uniform series, the factor equations can similarly be solved using logarithms—for example, from $F = A(F/A, i, n)$, one obtains $n = \ln(Fi/A + 1) / \ln(1 + i)$. The Excel function `=NPER(i, A, P, F)` provides a convenient alternative.

These tools are essential in practice: an investor may need to determine the *rate of return* earned on an investment, or a planner may need to know *how long* it takes for savings to reach a target amount.

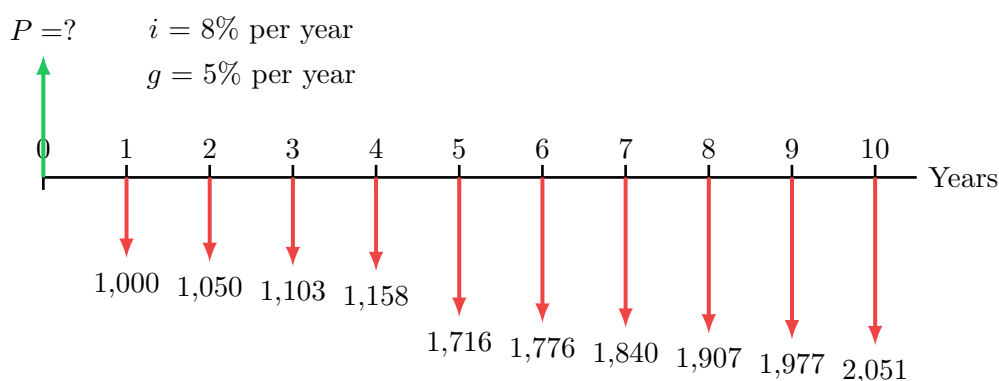
The cash flow diagram will allow a robust analysis of any series of cashflows that contain non-homogeneity. In the next chapter, we will build upon these concepts to analyse more complex cash flows and properly account for nominal versus effective interest rates.

Example 2.8. Composite Cash Flow: Geometric Gradient and Delayed Annuity

A company is upgrading its manufacturing facility. The upgrade has two cost components over a 10-year period:

1. Maintenance costs for a new machine start at \$1,000 at the end of Year 1 and increase by a geometric rate of 5% each year through Year 10.
2. An additional software subscription cost of \$500 per year begins at the end of Year 5 and continues through Year 10 (a total of 6 payments).

If the company's Minimum Attractive Rate of Return (MARR) is 8% per year, what is the total equivalent present worth (P) of these costs at Year 0?

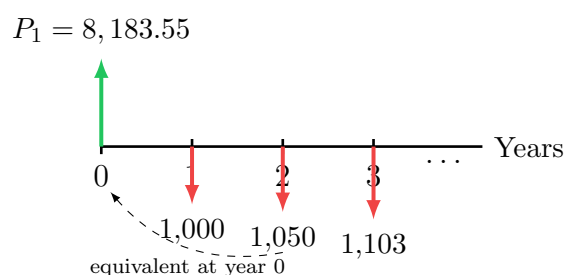
**Solution:**

The total present worth is the sum of the present worth of the geometric series (P_1) and the present worth of the delayed uniform series (P_2).

Step 1: Present Worth of the Geometric Gradient Series (P_1)

For the first component, we have $A_1 = 1,000$, $g = 0.05$, $i = 0.08$, and $n = 10$. Since $i \neq g$, we use the geometric gradient present worth factor:

$$\begin{aligned}
 P_1 &= A_1 \left(\frac{\left(\frac{1+g}{1+i} \right)^n - 1}{g - i} \right) \\
 &= 1,000 \left(\frac{\left(\frac{1+0.05}{1+0.08} \right)^{10} - 1}{0.05 - 0.08} \right) \\
 &= 1,000 \left(\frac{(0.9722)^{10} - 1}{-0.03} \right) \\
 &= 1,000 \left(\frac{0.7545 - 1}{-0.03} \right) \\
 &= 1,000(8.1836) \\
 &= 8,183.55
 \end{aligned}$$

**Step 2: Present Worth of the Delayed Uniform Series (P_2)**

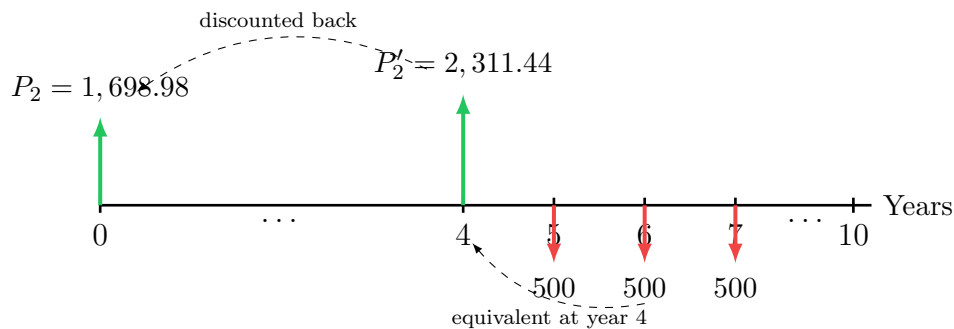
The second component is an annuity of $A_2 = 500$ starting at Year 5 and ending at Year

10. There are $10 - 5 + 1 = 6$ payments. First, we calculate the present worth of this annuity at Year 4 (one period before the first payment), which we denote as P'_2 :

$$\begin{aligned} P'_2 &= A_2(P/A, 8\%, 6) \\ &= 500 \left(\frac{(1 + 0.08)^6 - 1}{0.08(1 + 0.08)^6} \right) \\ &= 500(4.6229) \\ &= 2,311.44 \end{aligned}$$

Now, we discount P'_2 from Year 4 back to Year 0 using the single-amount present worth factor:

$$\begin{aligned} P_2 &= P'_2(P/F, 8\%, 4) \\ &= 2,311.44(1 + 0.08)^{-4} \\ &= 2,311.44(0.7350) \\ &= 1,698.98 \end{aligned}$$



Step 3: Total Present Worth (P)

The total equivalent present worth at Year 0 is the sum of the two components:

$$P = P_1 + P_2 = 8,183.55 + 1,698.98 = 9,882.53$$

2.8 Formula List

Table 2.5 summarizes all the factors, their standard notations, and their equations.

Name	Notation	Equation	Excel
Compound Amount	$(F/P, i, n)$	$(1 + i)^n$	$= FV(i\%, n, , -P)$
Present Worth	$(P/F, i, n)$	$(1 + i)^{-n}$	$= PV(i\%, n, , -F)$
Uniform Series Present Worth	$(P/A, i, n)$	$\frac{(1+i)^n - 1}{i(1+i)^n}$	$= PV(i\%, n, -A)$
Uniform Series Compound Amount	$(F/A, i, n)$	$\frac{(1+i)^n - 1}{i}$	$= FV(i\%, n, -A)$
Capital Recovery	$(A/P, i, n)$	$\frac{i(1+i)^n}{(1+i)^n - 1}$	$= PMT(i\%, n, -P)$
Sinking Fund	$(A/F, i, n)$	$\frac{i}{(1+i)^n - 1}$	$= PMT(i\%, n, , -F)$
Arithmetic Gradient Present Worth	$(P/G, i, n)$	$\frac{(1+i)^n - ni - 1}{i^2(1+i)^n}$	—
Arithmetic Gradient Uniform Series	$(A/G, i, n)$	$\frac{1}{i} - \frac{n}{(1+i)^n - 1}$	—
Geometric Gradient Present Worth ($i \neq g$)	$(P/A, g, i, n)$	$\frac{\left(\frac{1+g}{1+i}\right)^n - 1}{g - i}$	—
Geometric Gradient Present Worth ($i = g$)	$(P/A, g, i, n)$	$\frac{n}{1+i}$	—

Table 2.5: Summary of Engineering Economy Factors